higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

CIVIL AND STRUCTURAL STEEL WORK DETAILING

(First Paper)

NQF LEVEL 3

(12050663)

15 March 2019 (X-Paper)
09:00–11:00

REQUIREMENTS: **Hot-rolled structural steel sections (BOE 8/2)**
A4 drawing sheets

Calculators and drawing instruments may be used.

This question paper consists of 5 pages, 1 addendum and 1 answer sheet.

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TIME: 2 HOURS
MARKS: 75

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each question on a NEW page.
5. Rule off across the page on completion of each question.
6. Drawings and sketches must be done neatly using drawing instruments.
7. Write neatly and legibly.

NCV

QUESTION 1: GENERAL

Choose a description from COLUMN B that matches a term in COLUMN A. Write only the letter (A–O) next to the question number (1.1–1.15) in the ANSWER BOOK.

COLUMN A		COLUMN B	
1.1	Retaining wall	A	point at which a material begins to deform permanently
1.2	Formwork	B	made liquid by heating
1.3	Stirrup	C	thin steel bars welded together to make a crisscross pattern
1.4	Shutter	D	temporary structure or mould into which newly mixed concrete is placed and compacted to harden
1.5	Mesh	E	pieces of plate found at the intersection to facilitate a connection between
1.6	Yield strength	F	spacing between the centre lines of holes in the longitudinal direction of the member
1.7	Embedded	G	individual pieces of timber, plywood or steel used to make formwork
1.8	Butt welding	H	short length of angle or channel section welded or bolted to a steel member enabling it to be fastened to another member
1.9	Flux	I	beam manufactured from plate material and used for large pans
1.10	Molten	J	fixed or set securely or deeply into cement or concrete
1.11	Gusset plate	K	freestanding wall used to hold back soil
1.12	Rafter	L	welded joints in which pieces of metal are jointed without overlapping
1.13	Girder	M	type of reinforcing in beams that holds rebars in place and resists shear
1.14	Cleat	N	main beams supporting the roof
1.15	Pitch	O	substance used in soldering and brazing to make metals more fusible

(15 × 1) **[15]**

QUESTION 2: REINFORCED-CONCRETE DETAILING

2.1 Give THREE advantages of a bar-bending schedule. (3 × 2) (6)

2.2 List THREE types of steel reinforcement. (3)

2.3 Name FIVE types of reinforced-concrete detailing. (5)

2.4 Differentiate between *R* and *Y* steel types. (2)

[16]

QUESTION 3: STRUCTURAL STEEL WORK DETAILING

3.1 Name FOUR types of roof sheeting. (4)

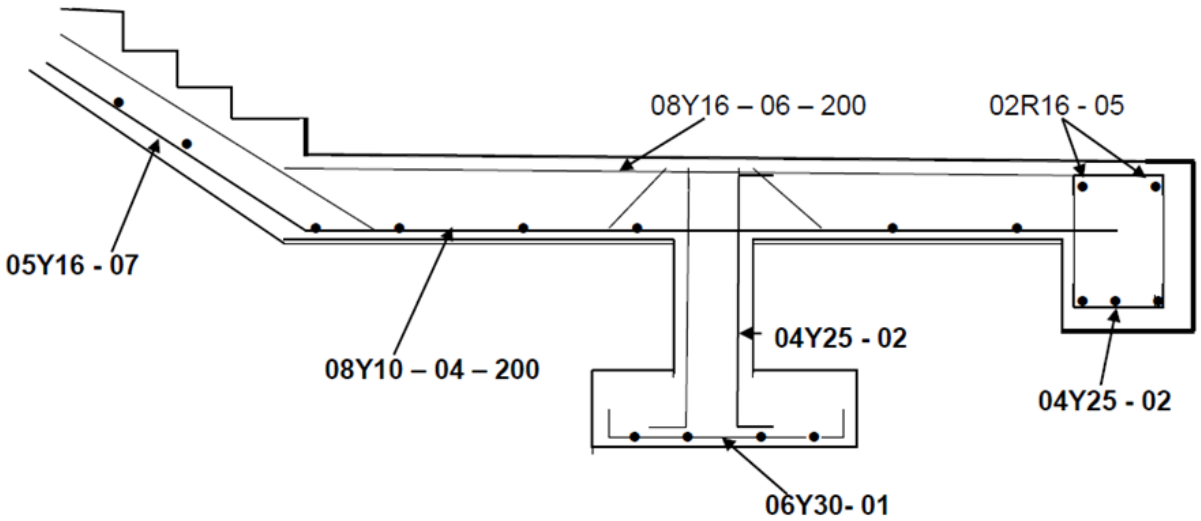
3.2 Make a freehand drawing of an N-type or Pratt truss lattice girder and label it. (5)

3.3 Give THREE advantages of portal-frame construction. (3 × 2) (6)

3.4 Shown below is a cross section through a concrete structure which includes a staircase, slab and pad foundation with a column and an edge beam.

Complete the steel schedule on the ANSWER SHEET (attached) using the given specifications.

NOTE: Ignore the length of each bar. A shape-code table is shown on the ADDENDUM (attached).



(14)

[29]

QUESTION 4: WELDING

4.1 Name FIVE types of welding joints. (5)

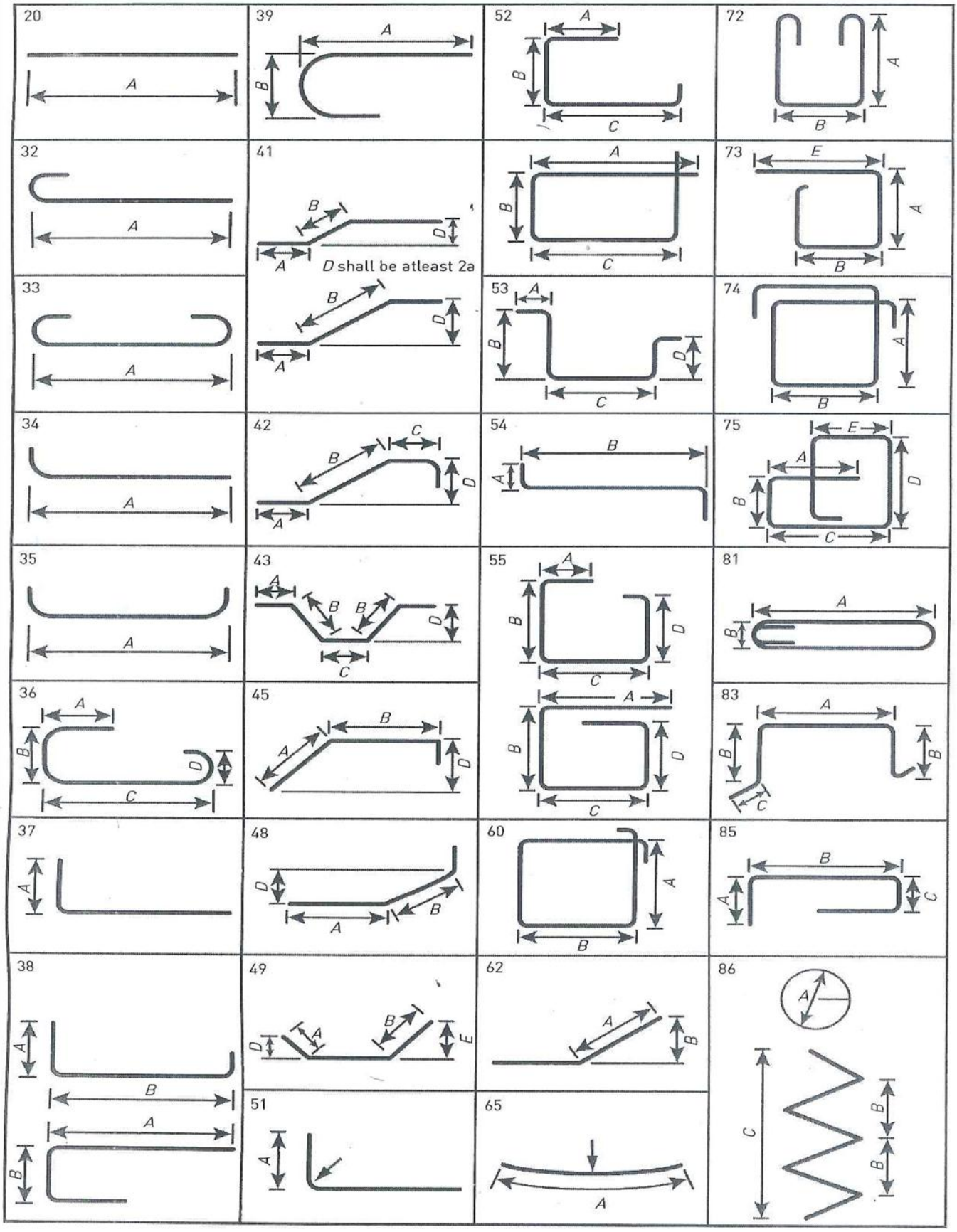
4.2 Draw simple sketches to illustrate each joint named in QUESTION 4.1. (5 × 2) (10)
[15]

TOTAL: 75

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ADDENDUM



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ANSWER SHEET EXAMINATION NUMBER:

Remove the ANSWER SHEET and attach it to the ANSWER BOOK.

3.4

1 Member	2 Bar mark	3 Type and size	4 No. of members	5 No. in each	6 Total number	7 Length of each bar	8 Shape

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